

An Analysis of Image super-resolution

Image super-resolution

Image super-resolution -- Part 1

Aliasing Geometrical SR reconstruction algorithms are possible if and only if the input low resolution images have been under-sampled and therefore contain aliasing. Because of this aliasing, the high-frequency content of the desired reconstruction image is embedded in the low-frequency content of each of the observed images. Given a sufficient number of observation images, and if the set of observations vary in their phase (i.e. if the images of the scene are shifted by a sub-pixel amount), then the phase information can be used to separate the aliased high-frequency content from the true low-frequency content, and the full-resolution image can be accurately reconstructed. In practice, this frequency-based approach is not used for reconstruction, but even in the case of spatial approaches (e.g. shift-add fusion), the presence of aliasing is still a necessary condition for SR reconstruction.

Image super-resolution -- Part 2

Some object features, though beyond the diffraction limit, may be known to be associated with other object features that are within the limits and hence contained in the image. Then conclusions can be drawn, using statistical methods, from the available image data about the presence of the full object. The classical example is Toraldo di Francia's proposition of judging whether an image is that of a single or double star by determining whether its width exceeds the spread from a single star. This can be achieved at separations well below the classical resolution bounds, and requires the prior limitation to the choice "single or double?" The approach can take the form of extrapolating the image in the frequency domain, by assuming that the object is an analytic function, and that we can exactly know the function values in some interval. This method is severely limited by the ever-present noise in digital imaging systems, but it can work for radar, astronomy, microscopy or magnetic resonance imaging. More recently, a fast single image super-resolution algorithm based on a closed-form solution to $\ell_2 - \ell_2$ problems has been proposed and demonstrated to accelerate most of the existing Bayesian super-resolution methods significantly.

Image super-resolution -- Part 3

Diffraction limit: the capacity of an optical instrument to reproduce the details of an object in an image has limits that are imposed by laws of physics: the diffraction equations in the wave theory of light, or the uncertainty principle for photons in quantum mechanics. Information transfer can never be increased beyond this boundary, but packets outside the limits can be cleverly swapped for (or multiplexed with) some inside it.

Super-resolution microscopy does not so much "break" as "circumvent" the diffraction limit. New procedures probing electro-magnetic disturbances at the molecular level (in the so-called near field) remain fully consistent with Maxwell's equations. Spatial frequency domain: A succinct expression of the diffraction limit is given in the spatial frequency domain. In Fourier optics light distributions are expressed as superpositions of a series of grating light patterns in a range of fringe widths - these widths represent the spatial frequencies. It is generally taught that diffraction theory stipulates an upper limit, the cut-off spatial-frequency, beyond which pattern elements fail to be transferred into the optical image, i.e., are not resolved. But in fact what is set by diffraction theory is the width of the passband, not a fixed upper limit. No laws of physics are broken when a spatial frequency band beyond the cut-off spatial frequency is swapped for one inside it: this has long been implemented in dark-field microscopy. Nor are information-theoretical rules broken when superimposing several bands, disentangling them in the received image needs assumptions of object invariance during multiple exposures, i.e., the substitution of one kind of uncertainty for another. Information: When the term super-resolution is used in techniques based on the inference of object details using a statistical treatment of the image within standard resolution limits (for example, averaging multiple exposures), it involves an exchange of one kind of information (extracting signal from noise) for another (the assumption that the target has remained invariant). Recent breakthroughs incorporate quantum-transformer hybrids into super-resolution, such as QUIET-SR, a 2025 model that employs shifted quantum window attention within a transformer to enhance image detail while respecting diffraction and information-theory limits. Similarly, frequency-integrated transformers (e.g., FIT) enrich super-resolution by explicitly combining spatial and frequency-domain information via FFT-based attention, improving reconstruction across scales. Resolution and localization: True resolution involves the distinction of whether a target, e.g. a star or a spectral line, is single or double, ordinarily requiring separable peaks in the image. When a target is known to be single, its location can be determined with higher precision than the image width by finding the centroid (center of gravity) of its image light distribution. The word ultra-resolution had been proposed for this process but it did not catch on, and the high-precision localization procedure is typically referred to as super-resolution.

Image super-resolution -- Part 4

Probing near-field electromagnetic disturbance Super-resolution microscopy is generally discussed within the realm of conventional optical imagery. However, modern technology

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allows the probing of electromagnetic disturbance within molecular distances of the source, which has superior resolution properties. See also evanescent waves and the development of the new super lens.

Image super-resolution -- Part 5

Multiplexing spatial-frequency bands An image is formed using the normal passband of the optical device. Then, some known light structure (for example, a set of light fringes) is superimposed on the target. The image now contains components resulting from the combination of the target and the superimposed light structure, e.g. moiré fringes, and carries information about target detail which simple unstructured illumination does not. The “superresolved” components, however, need disentangling to be revealed. For an example, see structured illumination (figure to left).

Image super-resolution -- Part 6

Known defects in a given imaging situation, such as defocus or aberrations, can sometimes be mitigated in whole or in part by suitable spatial-frequency filtering of even a single image. Such procedures all stay within the diffraction-mandated passband, and do not extend it.

Image super-resolution -- Part 7

Other related work Curtis, Craig H.; Milster, Tom D. (October 1992). "Analysis of Superresolution in Magneto-Optic Data Storage Devices". *Applied Optics*. 31 (29): 6272–6279. Bibcode:1992ApOpt..31.6272M. doi:10.1364/AO.31.006272. PMID 20733840. Zalevsky, Z.; Mendlovic, D. (2003). *Optical Superresolution*. Springer. ISBN 978-0-387-00591-1. Caron, J.N. (September 2004). "Rapid supersampling of multiframe sequences by use of blind deconvolution". *Optics Letters*. 29 (17): 1986–1988. Bibcode:2004OptL...29.1986C. doi:10.1364/OL.29.001986. PMID 15455755. Clement, G.T.; Huttunen, J.; Hynynen, K. (2005). "Superresolution ultrasound imaging using back-projected reconstruction". *Journal of the Acoustical Society of America*. 118 (6): 3953–3960. Bibcode:2005ASAJ..118.3953C. doi:10.1121/1.2109167. PMID 16419839. Geisler, W.S.; Perry, J.S. (2011). "Statistics for optimal point prediction in natural images". *Journal of Vision*. 11 (12): 14. doi:10.1167/11.12.14. PMC 5144165. PMID 22011382. Cheung, V.; Frey, B. J.; Jovic, N. (20–25 June 2005). *Video epitomes* (PDF). *Conference on Computer Vision and Pattern Recognition (CVPR)*. Vol. 1. pp. 42–49. doi:10.1109/CVPR.2005.366. Bertero, M.; Boccacci, P. (October 2003). "Super-resolution in computational imaging". *Micron*. 34 (6–7): 265–273. doi:10.1016/s0968-4328(03)00051-9. PMID

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